ARTICLE

PRISM: a modular system for glioma progression surveillance on MRI

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ARTICLE HISTORY

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ABSTRACT

Surveillance MRI after treatment for glioblastoma and other aggressive gliomas must be interpreted with prior imaging and the treatment course. A progression score alone does not show the clinician the evidence that produced it or what it leaves out. We built PRISM, a modular system that combines image-derived measurements, treatment context, separate image and timeline pathways, uncertainty and provenance in a locally generated, checked report. On MU-Glioma-Post, the timeline and image pathways reached AUCs of 0.82 and 0.71. Applied unchanged, the combined score reached 0.77 on LUMIERE; at UCSF-ALPTDG, single-scan image measurements reached 0.80 and the image pathway 0.73. The evidence behind the scores differed by cohort: treatment timing at MU-Glioma-Post and enhancing volume at UCSF-ALPTDG. In a second, cross-cohort report evaluation, 406 of 406 reports passed a lexical check against their structured records. PRISM is designed to make evidence inspectable, not to replace clinical judgement; its effect on clinical decisions needs a reader study.

KEYWORDS

glioma surveillance; external validation; clinical decision support

1. Introduction

Glioblastoma and other aggressive gliomas are followed with repeated MRI after surgery and chemoradiotherapy, and each scan is read against earlier scans and the course of treatment. The call is hardest soon after radiotherapy, when treatment can increase contrast enhancement without tumour growth. This pseudoprogression is most common in the first weeks after chemoradiotherapy, and more so when the promoter of the DNA-repair gene MGMT is methylated (Taal et al. 2008; Brandes et al. 2008).

A model's probability alone does not show the treatment context, the measurements behind it, its uncertainty or what it left out. Surveillance data make these gaps hard to close, because public datasets often hold images but few treatment dates, centres document progression differently, and the hardest situations are rare. Only 3 of 20 studies in a recent review of automated response assessment tested their models externally (Shi et al. 2025).

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We built PRISM to give the clinician, at each post-treatment scan, the image evidence and the treatment-timeline evidence separately, with a statement of how far each has been tested, in a report written inside the hospital. We developed it on MU-Glioma-Post (MU) and applied its models unchanged to LUMIERE and UCSF-ALPTDG (UCSF). The models ranked scans at both external centres, but what drove a score differed by cohort: the position in treatment at MU, and enhancement at UCSF.

2. Methods

2.1. PRISM

PRISM writes everything it computes for a scan, with the source of every value, into one structured record (Figure 1A; Table 1 names each score).

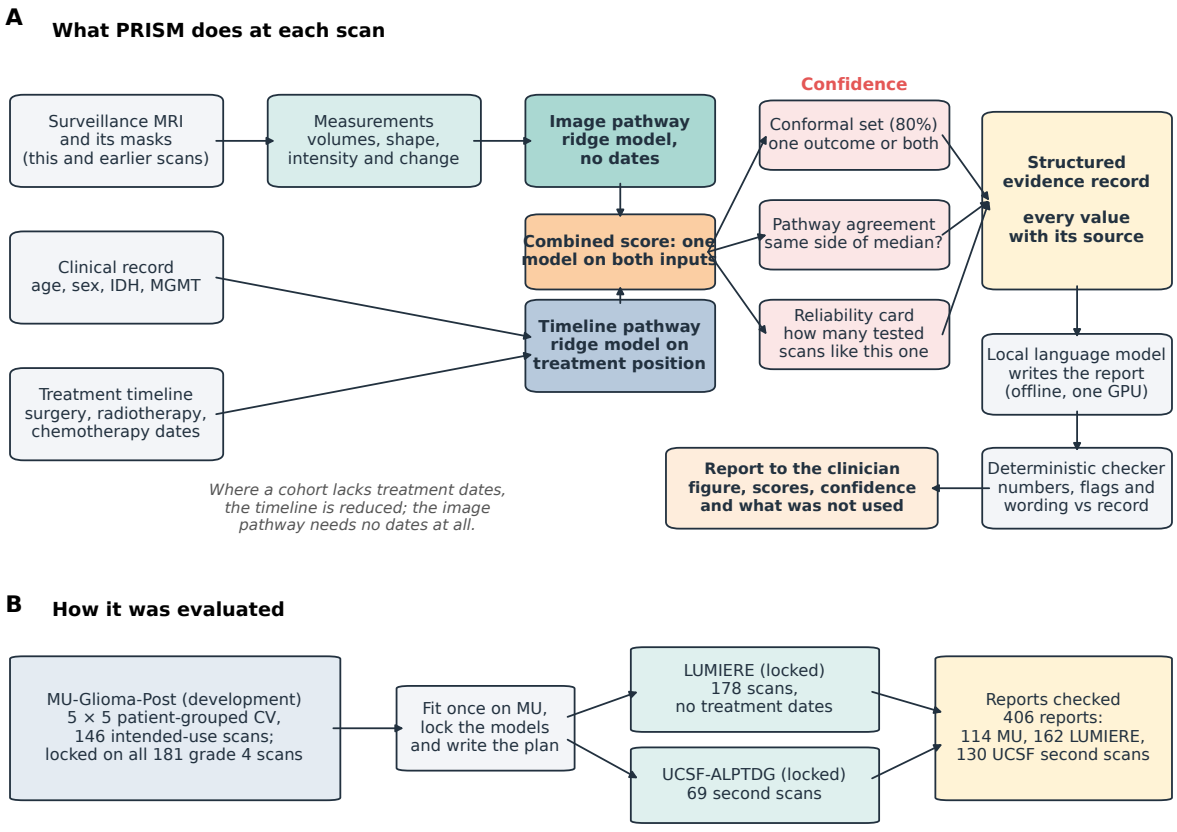


Figure 1. PRISM and how it was evaluated. (A) Measurements from the segmentation masks feed the image pathway; clinical data and the position in treatment feed the timeline pathway; one model on both sets of inputs gives the combined score. A three-part confidence is attached, every value enters one record with its source, and a local language model writes the report, which a deterministic checker tests against the record. (B) Models were developed on MU by repeated cross-validation, then fitted once and applied unchanged at LUMIERE and UCSF under plans committed in advance.

The timeline pathway scores a scan from clinical data and the patient's position in treatment, both taken as of the scan day. Where a cohort lacks treatment dates, a reduced timeline keeps the inputs it has (Table 1), with the flag for the first weeks after radiotherapy taken from the rater's note

Table 1. The scores built from measurements and records. Every score listed is a ridge logistic regression except enhancing volume alone; all are fitted within the training data, with unknown IDH or MGMT status as the reference category and missing measurements imputed with an indicator. The fine-tuned networks are described in Section 2.3. Status: planned before the analysis, or not fixed in advance (†).

Score	Inputs	Dates	Status
Clinical data alone	Age, sex, IDH and MGMT status	No	Planned reference (LUMIERE †)
Timeline pathway	Clinical inputs; days since diagnosis and surgery; radiotherapy start, end and dose; chemotherapy; visit index	Yes	Planned
Reduced timeline	Clinical inputs; days since surgery; flag for the first weeks after radiotherapy	Partly	Planned at UCSF-ALPTDG; † at LUMIERE
Image pathway	27 measurements of the segmented tumour: volumes, shape and intensity, change since the previous scan, volumes relative to the patient's smallest	No	† on MU; planned at UCSF-ALPTDG and in the system tests
Single-scan images	The 19 measurements of the current scan	No	† on MU; planned at UCSF-ALPTDG
Combined score	Inputs of both pathways; externally, the reduced timeline, volumes and their change, and the interval since a previous scan	Yes	† on MU; planned and locked at LUMIERE and UCSF-ALPTDG
Planned imaging model	Timeline inputs, image measurements and three schedule inputs (interval since the previous scan, whether there was one, smallest volume since the end of radiotherapy)	Yes	Planned primary model on MU; not part of PRISM
Enhancing volume alone	Log enhancing tumour volume, no fitted model	No	† reference

at LUMIERE and approximated at UCSF. The image pathway scores the scan from 27 measurements of the segmented tumour and its change (Table 1; [Supplementary section: image pathway inputs and their harmonisation at LUMIERE and UCSF]) and uses no dates or intervals. The measurements come from the masks released with each cohort (Mahmoud et al. 2025; Suter et al. 2022; Fields et al. 2024).

The confidence has three parts. A patient-level split conformal set at the 80% level (Angelopoulos and Bates 2023) holds one outcome when the score is clear and both when it is not; it is calibrated on 30% of the MU patients (within each training fold on MU) and applied unchanged elsewhere, which assumes that external scans are exchangeable with MU's ([Supplementary section: system results]). The two pathways agree when both scores lie on the same side of their medians over the cohort's evaluation scans; because those medians need a batch of scans, we also tested medians fixed on MU, which a single scan can use (exploratory). The reliability card (Mitchell et al. 2019) looks up the scan's situation, its phase of care crossed with the change in enhancing tumour, in a record of that cohort's held-out results, which includes the scan itself and so describes it slightly optimistically. A situation is tested with at least 10 scans of each outcome (the card gives AUCs), has few cases below that (counts only), and is not tested with none.

A local language model (Qwen3.8-27B, temperature 0) writes the report from the record alone, so patient data do not leave the hospital. A deterministic checker rejects any number, flag or confidence statement that is not in the record, any missing required content, and any wording that offers a diagnosis, advice or a probability (Supplementary Section S19).

2.2. Cohorts and the task

Grade 4 glioma is the primary population in all three cohorts (Table 2); we fitted each model once on MU and applied it unchanged, or locked, to the others. The task is to decide at each scan whether

it is the scan at which progression was documented, as each cohort documents it. On MU a positive is the first scan on or after the documented progression date, and a negative is a scan followed by at least 120 days without progression. At LUMIERE a positive is the first scan rated as progressive disease, and a negative is an earlier scan rated stable or responding. UCSF holds one pair of consecutive scans per patient, and we evaluated the second: a positive is a second scan whose progression date lies within 30 days of it and closer to it than to the first scan, and a negative is followed by at least 120 days without progression. On MU the intended use is scans from the start of radiotherapy, divided into three phases of care: during radiotherapy, early after it (up to 12 weeks after its end) and later; MU results are for the intended use unless stated. Predicting progression within 120 days of a scan is a secondary task.

2.3. Evaluation

The locked models were fitted on all 181 grade 4 detection scans of MU, from every phase of care, and the external tests followed plans committed before any score was compared with an external label. The one deviation, in which MU scans the UCSF plan named for fitting, left the LUMIERE AUC almost unchanged ([Supplementary section: deviations from the plans]). On MU we evaluated every model by five repeats of five-fold cross-validation grouped by patient and report the mean pooled held-out AUC with a 95% confidence interval (CI) from 1,000 patient-level bootstrap draws, as for every external AUC; no patient contributed to more than one of fitting, calibration and testing. We used repeated splits because single splits are unstable in small cohorts (An et al. 2021; Varoquaux 2018): an earlier radiomics model on MU reached 0.80 on one test split but averaged 0.72 when re-selected and re-tested over 500 splits of the same design (Supplementary Section S8 and Figure S2). As references, we fitted each UCSF model on UCSF by cross-validation, and fine-tuned segmentation networks to score scans directly against a permuted-label control ([Supplementary section: deep networks at MU and UCSF]). Results of analyses not fixed in advance carry a dagger (†); the disagreement flag and its tests were planned, but its median rule was written down after the results had been seen ([Supplementary section: analyses not fixed in advance]).

3. Results

3.1. The timeline carries most of the evidence at the development centre

In the intended use on MU (146 scans), the timeline pathway detected the progression scan with an AUC of 0.82 (95% CI 0.75–0.90) and the image pathway with 0.71 (0.63–0.78)†; clinical data alone scored 0.53 (0.45–0.61; Figure 2). The combined score also reached 0.82 (0.74–0.89)†, so the images did not raise the timeline's AUC, and a planned model adding image measurements and three schedule inputs to the timeline fell short of its target gain of 0.03 (Table 3). Most of the timeline's advantage lay between phases of care (Section 3.3). Forecasting progression within 120 days was harder, at 0.60 (0.52–0.69) in the intended use (Supplementary Section S23).

Table 2. The three cohorts. What each cohort records, and the scans in the detection task. Grade 4 glioma is the primary population.

	MU-Glioma-Post	LUMIERE	UCSF-ALPTDG
Role	Development (repeated cross-validation)	External, locked	External, locked
Access	Public	Public	Data use agreement
Scans per patient	Up to 6 scans with a 120-day label (median 2)	Up to 10 scans with a 120-day label (median 2)	One consecutive pair
Treatment dates	Surgery, radiotherapy start and end, dose, chemotherapy	Days from surgery only; rater flag for scans within three months of radiotherapy	Days from surgery; radiotherapy start only
Documented progression (the label)	Documented date in the clinical record	Expert response rating of each scan	Progression date supplied with the dataset
Segmentation masks	Automatic masks released with the dataset	DeepBraTumIA (three labels)	Masks provided with the dataset
Patients[a]	126	61	140
Age, median	62.5 (IQR 53–70)	62 (IQR 53–70)	57.5
Male	66 (52%)	32 (52%)	81 (58%)
IDH wildtype / mutant / unknown	112 / 8 / 6	38 / 1 / 22	81 / 13 / 46
MGMT methylated / unmethylated / unknown	45 / 62 / 19	26 / 28 / 7	71 / 43 / 26
Detection task[b]: progression / non-progression scans (patients)	62 / 84 (80) in the intended use; 64 / 117 (92) over all scans	73 / 105 (79)	29 / 40 (69), second scan only
Phase of care in the detection set	Progression: 56 of 62 later after radiotherapy; non-progression: 51 of 84 early after radiotherapy	Not recorded	65 of 69 later after radiotherapy[c]

[a] Patient characteristics describe the cohorts of the secondary 120-day task at MU-Glioma-Post (grade 4) and LUMIERE, and all grade 4 patients at UCSF-ALPTDG. Age is at diagnosis at MU-Glioma-Post and at surgery at LUMIERE.

[b] Positives are the scan at which progression was documented; negatives are defined per cohort in Section 2.2. The LUMIERE detection set has more patients than the 120-day cohort because it includes patients who contribute only a progression scan. At MU-Glioma-Post, the 146 scans of the intended use are the 181 grade 4 detection scans less the 35 taken before radiotherapy started, and 42 grade 4 progression scans of patients with no scan carrying a 120-day label are excluded. Reports were evaluated on the 114 intended-use scans of the 60 MU-Glioma-Post patients not used to develop the prompt, on 162 of the 178 LUMIERE scans and on 130 of the 140 grade 4 second scans at UCSF-ALPTDG, the rest belonging to development patients (Section 3.5).

[c] UCSF-ALPTDG gives only the first radiotherapy start; the phase assumes a six-week course. Two scans were early after radiotherapy and two had no radiotherapy date.

Table 3. Questions fixed in advance and their answers. Each rule was committed to the repository before the analysis it governs. AUC differences with 95% CIs.

Question	Rule fixed in advance	Result	Met
Detection on MU-Glioma-Post			
Do image measurements add to the timeline?	Planned model ≥ 0.75 (lower bound ≥ 0.65) and gain ≥ 0.03 with CI above 0	Model 0.837 (0.763–0.904); gain +0.015 (−0.032 to +0.063)	No
Do fine-tuned deep networks add?	Gain ≥ 0.03 with CI above 0; permuted-label control covers 0.5	No network added; control 0.483 (0.436–0.527)	No (control valid)
Does the locked model transfer to LUMIERE?	External AUC ≥ 0.65	AUC above 0.65 (Section 3.2)	Yes
UCSF-ALPTDG			
Does the image pathway transfer?	AUC ≥ 0.65, lower bound > 0.5	Both conditions met (Section 3.2)	Yes
Do images add to clinical data without a timeline?	Gain with CI above 0	+0.248 (+0.096 to +0.392)	Yes
Does training at UCSF-ALPTDG do better than the locked model?	Descriptive	Image pathway higher when trained there; single-scan model about the same (Section 3.2)	Descriptive
Do deep networks learn at UCSF-ALPTDG?	Above the permuted-label control; control covers 0.5	+0.201 (+0.094 to +0.306) over a control of 0.541 (0.480–0.603)	Yes
The system (MU-Glioma-Post, LUMIERE, UCSF-ALPTDG)			
Do conformal sets keep their coverage?	Coverage ≥ 0.78 at the 80% level in each cohort	Combined score met on point estimates in all three, but not for every outcome (Section 3.4); images at LUMIERE 0.708 and timeline at UCSF-ALPTDG 0.696	Yes (combined)
Do flags pick out less accurate scans?	AUC unflagged minus flagged > 0, CI above 0	Disagreement at MU-Glioma-Post +0.307 (+0.175 to +0.432); per-cohort results in the supplement[a]	MU: disagreement and set; LUMIERE: set; UCSF: neither
Does setting flagged scans aside beat matched removal?	Retained AUC minus matched removal > 0, CI above 0	Disagreement met at MU-Glioma-Post (Section 3.4); per-cohort results in the supplement[a]	MU: disagreement; LUMIERE: neither; UCSF: set
Do tested situations have lower error?	Descriptive: tested below few cases in Brier score and error	Brier 0.187 vs 0.258 at LUMIERE; reversed at MU-Glioma-Post (Section 3.4)	Descriptive
Are reports faithful to the record?	≥ 95% pass the checker in each cohort	First evaluation failed at LUMIERE; second evaluation: all passed (Section 3.5)	No, then yes
Is the tier stated where evidence is thin?	100%	All (Section 3.5)	Yes
Do reports follow a changed record?	≥ 27 of 30 perturbed records	28 of 30 (Section 3.5)	Yes
Smallest report model passing the checker rule in every cohort	Descriptive	Qwen3.8-27B (supplement)	Descriptive

Plans were committed before each analysis: detection (commit 2b4c927), UCSF-ALPTDG (0e7c9fe), the system (e7cba80) and the second report evaluation (4f7e7b9). In the first row, the planned model gained +0.041 (+0.006 to +0.081) over all scans of all gliomas, which may reflect its schedule inputs (Section 3.3). Matched removal keeps the phase-of-care mix of the removed scans (random removal at LUMIERE, which has no phase); "set" is a conformal set holding both outcomes. The flag and tier tests were fixed in advance, but the agreement rule and the card tiers were defined after the cohort results had been seen†.

[a] The Met cells use each cohort's own medians, the rule the reports use. With medians fixed on MU (Section 2.1), disagreement also met both tests at LUMIERE†. [Supplementary section: system results]: both tests for each flag in each cohort, with both median rules and with the rule that flags any scan.

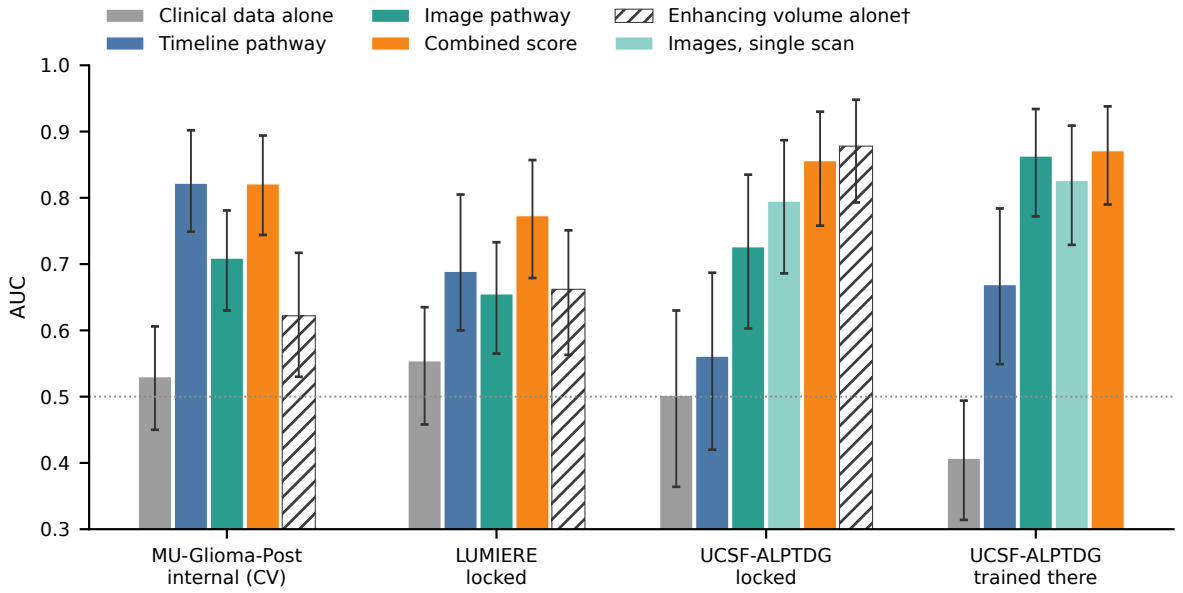


Figure 2. The models rank scans at two external centres without refitting. AUC for detecting the scan at which progression was documented. MU: repeated cross-validation, intended use. LUMIERE and UCSF: models fitted once on MU and applied unchanged, with the reduced timeline. UCSF trained there: reference. At UCSF the images are shown for the single-scan model and for the image pathway with change inputs, the system's image pathway; the clinical-only model at LUMIERE was not fixed in advance†. Hatched: log enhancing tumour volume alone†. Error bars, 95% patient-bootstrap CIs; dotted line, chance.

3.2. The models rank scans at two external centres without refitting

The locked combined model, whose external form scored 0.80 (0.73–0.89) on MU by cross-validation, reached 0.77 (0.68–0.86) at LUMIERE and met its planned transfer rule (Table 3), above the image pathway and the reduced timeline† (Figure 2), although its gain over the timeline had an interval that included zero. At UCSF both planned image models transferred: the single-scan measurements reached 0.80 (0.69–0.89) and the image pathway 0.73 (0.60–0.84). With coefficients fitted on MU, the image pathway scored 0.07 below the single-scan model (−0.13 to −0.01). Trained at UCSF, it gained 0.14 over its locked version (0.03–0.26), while the single-scan model stayed about the same†. The reduced timeline scored 0.56 (0.42–0.69) at UCSF, where its radiotherapy flag is approximated, and the locked combined score reached 0.86 (0.76–0.93), partly from the interval between the two scans, which was shorter before progression†.

Read as a risk, the combined score was calibrated in the large at LUMIERE† but overpredicted progression at UCSF (observed over expected 0.72, 0.55–0.91), and a threshold at the MU median† called 90% of UCSF scans positive ([Supplementary section: calibration and thresholds]).

3.3. What carries a score differs by cohort

At MU the scores depended mainly on where the patient was in treatment, and at UCSF on how much the tumour enhanced† (Figure 3). At MU most progression scans came more than 12 weeks after radiotherapy and most non-progression scans within them (Table 2), so the schedule alone separates most of them. A within-phase AUC pools only the pairs of a progression and a non-progression scan from the same phase of care. Adding the timeline to clinical data and images gained 0.11 (0.04–

0.20) across phases of care and −0.02 (−0.13 to +0.11) within them (Figure 3A). Within a phase, the image pathway scored 0.68 (0.58–0.77) and the timeline 0.64 (0.51–0.78).

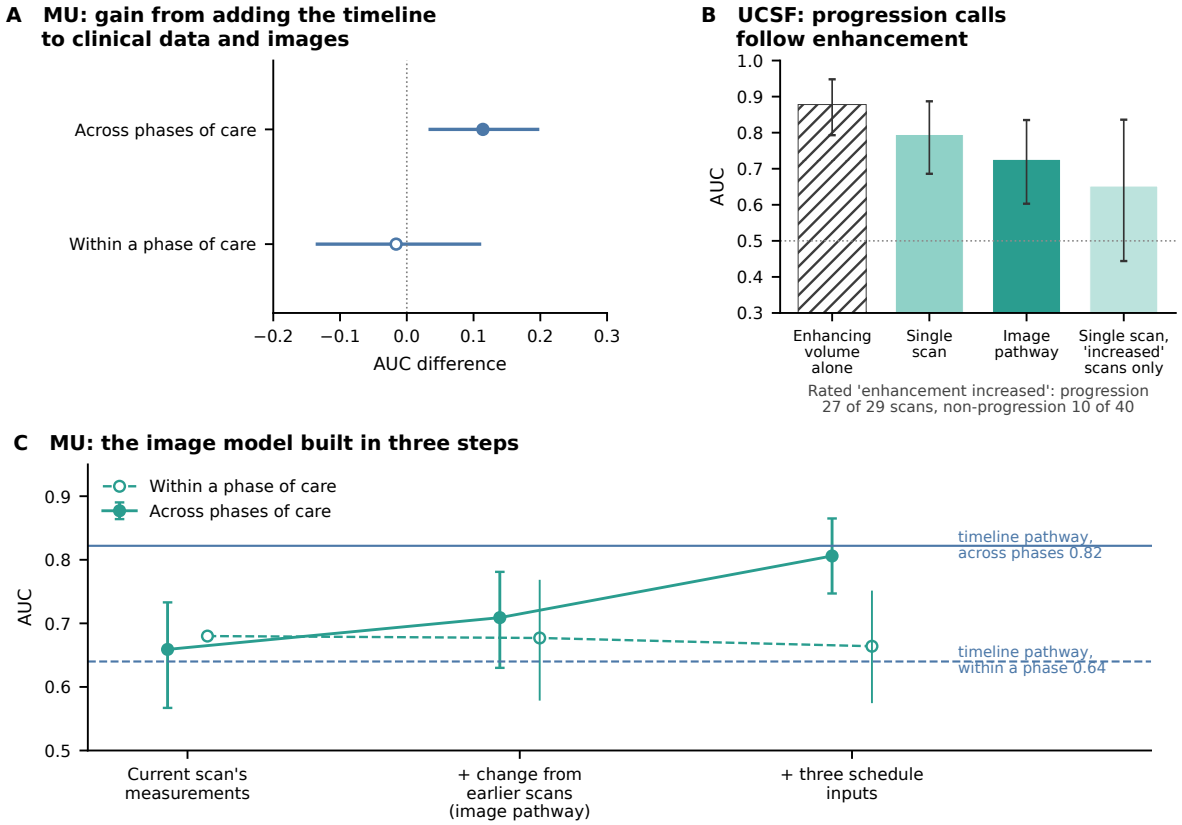


Figure 3. What carries a score differs by cohort. (A) MU: gain from adding the timeline to clinical data and images, across and within phases of care. (B) UCSF: enhancing volume alone beside the locked image models; the last bar is restricted to the 37 scans rated as showing increased enhancement. (C) MU: the image model built in three steps, across and within phases of care, against the timeline pathway; the interval of the second step includes zero. The audits were not fixed in advance (†); the locked UCSF AUCs in (B) were planned. Error bars, 95% patient-bootstrap CIs.

On MU the image model gained most from inputs that carry timing†. Adding change from earlier scans to the current scan's measurements gave the image pathway, 0.05 higher with an interval that included zero (−0.003 to +0.106). Adding the planned model's three schedule inputs (Table 1) raised the AUC a further 0.10 (0.05–0.15) across phases of care but not within them (−0.01, −0.06 to +0.04) (Figure 3C).

UCSF rated enhancement as increased at 27 of the 29 progression scans, and progression was dated a median of 7 days after the scan (Figure 3B). Enhancing volume alone† reached 0.88 (0.79–0.95), above the point estimates of the locked image models. Among the 37 scans rated as increased, the single-scan model reached 0.65 (0.44–0.84), so beyond that rating the measurements added little that this sample could detect. Deep networks fine-tuned on UCSF scans beat a permuted-label control (Table 3), and their scores tracked enhancing volume† ([Supplementary section: deep networks at MU and UCSF]).

3.4. Coverage holds on average; disagreement marks the less accurate scans at MU

The 80% conformal sets of the combined score covered 78% to 87% of scans in the three cohorts, at or above the planned threshold of 78%, and held a single outcome for 62% to 75% of scans (Figure 4A). At UCSF the interval reached below the threshold (68% to 87%). Coverage was uneven by outcome†: 77% (67% to 87%) for progression scans at LUMIERE and 65% (51% to 79%) for non-progression scans at UCSF, against 94% and 97% for the other outcome. Calibrating the sets separately for each outcome did not correct this.

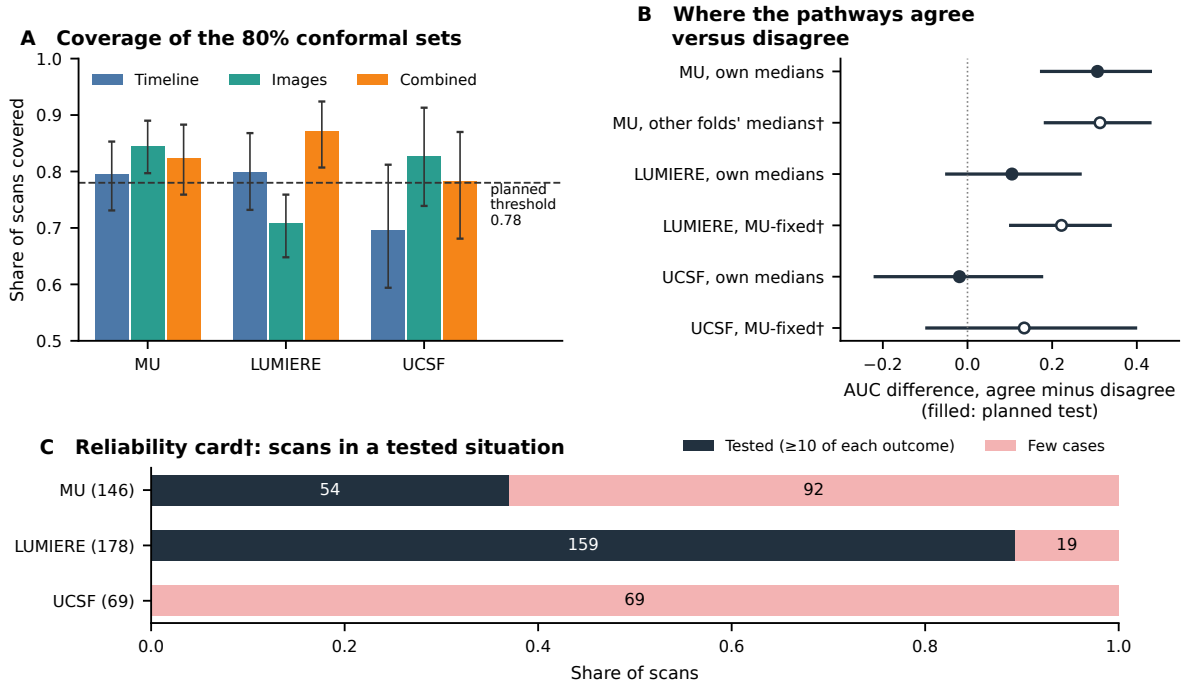


Figure 4. The confidence keeps its coverage on average; disagreement marks the less accurate scans at MU. (A) Coverage of the 80% conformal sets by cohort and pathway; dashed line, the planned threshold. (B) The combined score's AUC where the pathways agree minus where they disagree, with each cohort's own medians (filled, planned) and with medians fixed on MU (open†; on MU, the other folds' medians). (C) Scans whose situation reached the reliability card's† tested tier; the rest had few cases.

Disagreement between the pathways marked the less accurate scans at MU but not, under the planned rule, at the external centres. On MU the pathways disagreed on 40% of scans; the combined score's AUC was 0.31 higher where they agreed (0.18 to 0.43), and setting the disagreeing scans aside gained 0.08 (0.05–0.12) more than removing as many scans matched for phase of care, meeting both planned tests (Table 3; Figure 4B). With each cohort's own medians, the rule the reports use, neither planned test was met at LUMIERE or UCSF (at LUMIERE, 0.11, −0.05 to 0.27). With medians fixed on MU†, disagreement flagged 60% of LUMIERE scans and met both tests there (AUC difference 0.22, 0.10–0.34; gain over random removal 0.12, 0.05–0.19), but not at UCSF. A conformal set holding both outcomes, used as a flag, met one of the two planned tests in each cohort.

Only four situations on the reliability card† reached the tested tier, one at MU and three at LUMIERE, where no phase of care is recorded. Every UCSF situation had few cases, and at MU no situation in the first 12 weeks after radiotherapy held more than three progression scans (Figure 4C). Tested situations had lower error than few-case ones at LUMIERE, with overlapping intervals; at

MU the order was reversed, because the few-case scans were mostly early non-progression scans that the models score easily (Table 3).

3.5. After one revision, every report passed an automatic check against its record

Every report passed the check against its record: 114 at MU, 162 at LUMIERE and 130 at UCSF (Figure 5). All 220 reports in few-case or untested situations said so, and 28 of 30 reports followed a changed field, against a planned minimum of 27 (Table 3). This was after one revision of the prompt†: before it, every LUMIERE report failed, because a fixed sentence quoted a number held in no field; we corrected the prompt, kept the checker and evaluated again on patients not used for the revision ([Supplementary section: report revision]). The check is lexical and cannot judge emphasis or what a clinician would infer.

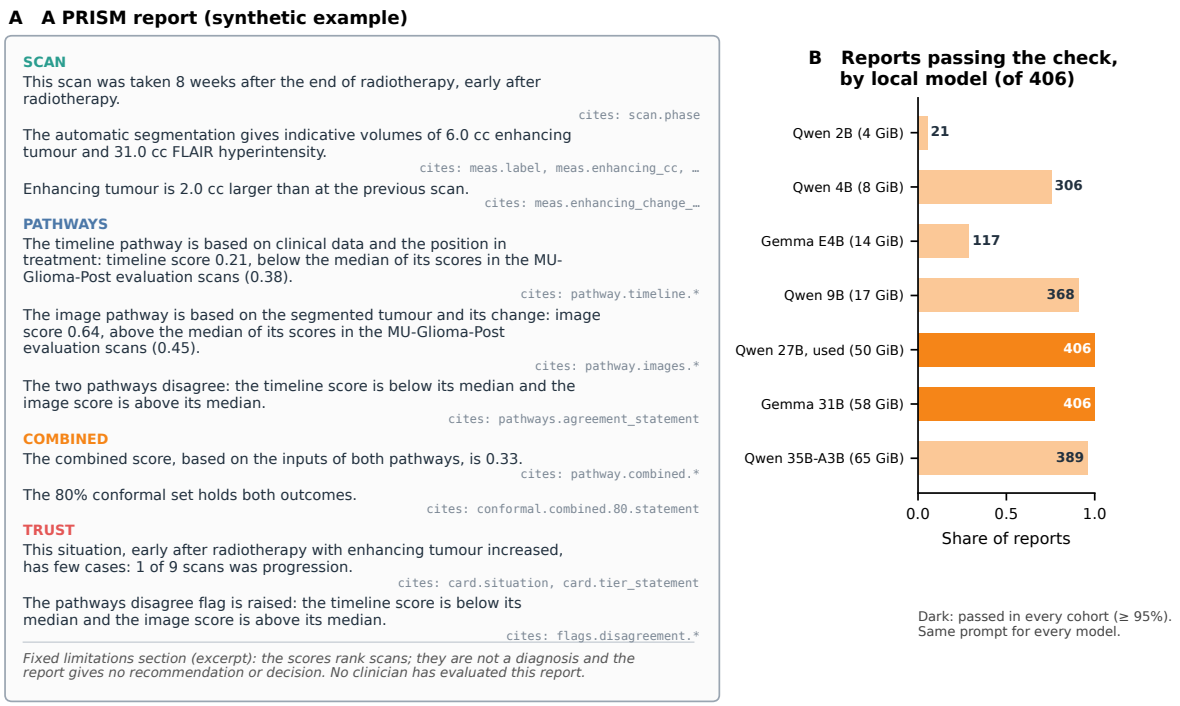


Figure 5. What the clinician receives, and the check. (A) The narrative of a report for a synthetic scan eight weeks after radiotherapy with growing enhancement (Section 4); volumes and scores are invented, and the card row is MU's. Each sentence lists the record fields it cites. The delivered report also holds the segmentation figure, tables of every value and a fixed limitations section. (B) Reports passing the check, of 406, for seven local language models given the same prompt (Qwen3.5 2B, 4B and 9B; Gemma-4 E4B and 31B; Qwen3.8-27B, used in PRISM; Qwen3.5-35B-A3B), after one revision of the prompt† (Section 3.5); dark bars passed in every cohort, and the model weights' size is in brackets.

4. Discussion

PRISM's models ranked scans at two other centres without refitting, but what carried a score depended on how each cohort was scanned and labelled. An AUC cannot show which evidence a model used; the separate pathways and the audits of Section 3.3 can. At MU it was mostly when

patients were scanned, and within a phase of care the images gave about as much as the timeline†; at UCSF it was the enhancement on which progression was called. The risk scale did not transfer to UCSF, so a new centre would recalibrate the score and set any threshold on local scans.

Because its modules are separate, PRISM runs on what a cohort holds: the image pathway needs only scans and masks, which most public datasets have; the timeline is added where dates exist; and the reliability card is a lookup table to which a new cohort adds rows, so a situation's tier can change as evidence accumulates.

Consider a scan eight weeks after radiotherapy in which the enhancing tumour has grown (Figure 5A). The timeline pathway scores such a scan low, because most scans at this point in treatment are not progression, and the image pathway scores it high, because enhancement has grown. The report shows both, states that they disagree and that the conformal set holds both outcomes, and so states that PRISM has too few tested cases to support a call here, in the weeks when pseudoprogression is most common, and the clinician decides with information PRISM does not use, such as symptoms, steroid dose and the radiotherapy plan.

Each component has been described before: longitudinal segmentation with measurement of change (Rudie et al. 2022), conformal prediction for glioma segmentation (Elyassirad et al. 2025), fusion of clinical and imaging data against pseudoprogression (Gomaa et al. 2025) and model cards (Mitchell et al. 2019). The closest system assigns a structured follow-up category from treatment variables extracted by a language model and tumour volumes, at one centre and without an uncertainty (Jabal et al. 2026). On these cohorts, earlier work predicted a patient-level label on MU (Christodoulou et al. 2026) and response categories on LUMIERE ([cite Tikhonov 2025, arXiv:2509.06511; full reference not yet verified]); separately, Chen et al. (2026) predicted recurrence risk from preoperative and postoperative MRI across centres. None detects progression at the scan, and we are not aware of a system that combines these components with each kind of evidence kept apart and the system's own test results attached.

PRISM was developed at one centre, and the two external tests are its only evidence that the ranking transfers. The label is progression as each centre documented it; no scan was re-read, and at UCSF most progression scans were also rated as showing increased enhancement. Of the confidence, only average coverage held at all three centres: under the planned rule, disagreement marked the less accurate scans at MU alone, and no UCSF scan fell in a tested situation. Subgroups are small, the analyses marked † need testing on new data, and no clinician has yet read the reports. PRISM forecast progression within 120 days less well than it detected progression at the scan (AUC 0.60, 0.52–0.69, in the intended use).

The next steps are to pool the rare situations (early progression after radiotherapy, and growing enhancement without progression) across cohorts, including the Burdenko and UCSD-PTGBM datasets, and test models fine-tuned on them there; and a reader study to show whether separate evidence and a stated confidence change clinicians' decisions.

Conclusion

PRISM keeps image and treatment-timeline evidence apart and shows the clinician both scores at every scan with how far each has been tested there. Its models ranked scans at two other centres without refitting, but what carried a score differed by cohort, which is why the evidence is shown apart.

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[PENDING AUTHOR DECISION].

Disclosure statement

[PENDING AUTHOR DECISION: competing interests, or "The authors report there are no competing interests to declare."] If any author serves on the editorial or advisory board of this journal, state it here.

Declaration of generative AI use

[DRAFT, PENDING AUTHOR CONFIRMATION:] Generative AI is part of the system studied. PRISM's report generator uses a local language model (Qwen3.8-27B; six other open-weight models in a comparison); all ran offline on the authors' computing cluster and saw only derived system records. An AI assistant (Claude, Anthropic) was used to write and run analysis code and to draft and edit the manuscript under the authors' direction; it was given no patient-level data through any hosted service. The authors reviewed all code, results and text and take full responsibility for the content of this article.

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Ethics statement

[PENDING AUTHOR DECISION: secondary analysis of de-identified datasets (MU-Glioma-Post and LUMIERE, public; UCSF-ALPTDG under a data use agreement); state the institutional determination or waiver and cite the source datasets' approvals.] No registered protocol. Analysis plans were committed to the repository before each external test; their commits are listed in Table 3. Patient and public involvement: none.

Data availability statement

[PENDING AUTHOR DECISION: MU-Glioma-Post (TCIA, doi:10.7937/7K9K-3C83) and LUMIERE (figshare; Suter et al. 2022) are public; UCSF-ALPTDG is available from its provider under a data use agreement; state which derived tables, fold assignments and locked model files are released, with a DOI.] Code: [PENDING: public repository link and release tag.]

Supplemental material

Supplemental data for this article are provided as a separate file.

Notes on contributors

[PENDING: a short biographical note for each author (no more than 200 words each).]

References

An C et al. 2021. Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results. PLOS ONE. 16(8):e0256152. https://doi.org/10.1371/journal.pone.0256152

Angelopoulos AN, Bates S. 2023. Conformal prediction: A gentle introduction. Found Trends Mach Learn. 16(4):494–591. https://doi.org/10.1561/2200000101

Brandes AA, Franceschi E, Tosoni A et al. 2008. MGMT promoter methylation status can predict the incidence and outcome of pseudoprogression after concomitant radiochemotherapy in newly diagnosed glioblastoma patients. J Clin Oncol. 26(13):2192–2197. https://doi.org/10.1200/JCO.2007.14.8163

Chen C, Guo F, Huang D et al. 2026. Predicting recurrence risk of glioblastoma based on preoperative-postoperative longitudinal MRI: A multicenter study. Bioengineering. 13(6):668. https://doi.org/10.3390/bioengineering13060668

Christodoulou RC, Vamvouras G, Pitsillos R, Solomou EE, Georgiou MF. 2026. Explainable radiomics with probability calibration for postoperative glioblastoma surveillance. Eur J Radiol Artif Intell. 5:100074. https://doi.org/10.1016/j.ejrai.2026.100074

Elyassirad D, Gheiji B, Vatanparast M, Ahmadzadeh AM, Faghani S. 2025. CONSeg: Voxelwise uncertainty quantification for glioma segmentation using conformal prediction. Am J Neuroradiol. 46(12):2553–2560. https://doi.org/10.3174/ajnr.A8914

Fields BKK et al. 2024. The University of California San Francisco adult longitudinal post-treatment diffuse glioma MRI dataset. Radiol Artif Intell. 6(4):e230182. https://doi.org/10.1148/ryai.230182

Gomaa A, Huang Y, Stephan P et al. 2025. A self-supervised multimodal deep learning approach to differentiate post-radiotherapy progression from pseudoprogression in glioblastoma. Sci Reports. 15:17133. https://doi.org/10.1038/s41598-025-02026-7

Jabal MS et al. 2026. Agentic automation of BT-RADS scoring: End-to-end multi-agent system for standardized brain tumor follow-up assessment. [Preprint, arXiv:2603.21494]. `https://arxiv.org/abs/2603.21494`

Mahmoud E, Gass J, Dhemesh Y et al. 2025. MU-Glioma Post: A comprehensive dataset of automated MR multi-sequence segmentation and clinical features. Sci Data. 12(1):1847. https://doi.org/10.1038/s41597-025-06011-7

Mitchell M, Wu S, Zaldivar A et al. 2019. Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). ACM. p 220–229

Rudie JD, Calabrese E, Saluja R et al. 2022. Longitudinal assessment of posttreatment diffuse glioma tissue volumes with three-dimensional convolutional neural networks. Radiol Artif Intell. 4(5):e210243. https://doi.org/10.1148/ryai.210243

Shi T, Kujawa A, Linares C, Vercauteren T, Booth TC. 2025. Automated longitudinal treatment response assessment of brain tumors: A systematic review. Neuro-Oncology. 27(8):1946–1971. https://doi.org/10.1093/neuonc/noaf037

Suter Y, Knecht U, Valenzuela W et al. 2022. The LUMIERE dataset: Longitudinal glioblastoma MRI with expert RANO evaluation. Sci Data. 9(1):768. https://doi.org/10.1038/s41597-022-01881-7

Taal W, Brandsma D, de Bruin HG et al. 2008. Incidence of early pseudo-progression in a cohort of

malignant glioma patients treated with chemoirradiation with temozolomide. Cancer. 113(2):405–410. https://doi.org/10.1002/cncr.23562

Varoquaux G. 2018. Cross-validation failure: Small sample sizes lead to large error bars. NeuroImage. 180:68–77. https://doi.org/10.1016/j.neuroimage.2017.06.061